

Kevin Du

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PROFESSIONAL EXPERIENCE

Software Engineer

Jan 2023 - Present

Optum

Backend engineer building APIs, microservices, AI agentic workflows, and test automation for a large-scale healthcare provider onboarding platform.

- Designed, built, and deployed a production ReAct-based AI agent on GCP using LangGraph/Google ADK and GPT models to execute multi-step reasoning over structured healthcare and insurance data.
- Developed and deployed 20+ RESTful APIs across 12 services using Java Spring Boot and Hibernate/JPA, supporting provider onboarding workflows for 10,000+ healthcare providers and handling approximately 500K requests monthly.
- Modernized a legacy Spring Boot platform by integrating AI agents to automate normalization of healthcare data into FHIR-compliant canonical models for downstream processing.
- Developed a Python-based test automation framework for CI/CD pipelines covering 2,000+ test scenarios, eliminating days of manual testing and reducing regression test execution to approximately 30 minutes.
- Improved application security by remediating 100+ vulnerabilities, migrating 12 Docker images to Chainguard hardened images, and integrating SonarQube into CI/CD pipelines to detect vulnerabilities before deployment.
- Built comprehensive unit, integration, load, and performance test suites using JUnit, Mockito, Pytest, Playwright, and JMeter to improve application reliability and identify performance bottlenecks.

Software Engineer Intern

Oct 2022 - Jan 2023

Aspen Dental Management Inc.

Back-end development & API/UI automated testing on the Customer and Practice Facing UI team.

- Implemented automation frameworks, increasing test coverage and reliability.
- Developed and tested API's for scheduling clinical procedures.
- Developed and evolved test plans for complex applications with high reliability and relevancy.

Software Engineer Intern

Feb 2022 – Oct 2022

Zoro US

Back-end development on the Quality Engineering and UI Automation team.

- Developed and implemented UI & API tests to enhance code coverage and strengthen the overall test coverage within the test library using Cypress & Playwright.
- Enhanced automated test frameworks for increased test throughput and execution flexibility.
- Developed features for our internal tool used to automate processes that ensure efficiency of our services.

TECHNICAL SKILLS

Languages: Python, Java, HTML/CSS, JavaScript, SQL, Go

Tools/Frameworks: Cypress, Selenium, Pytest, Spring Boot, FastAPI, Playwright, Node.js

Technologies: Git, Github Actions, Jenkins, GCP, AWS, S3, Kubernetes, Docker, Kafka, MongoDB, Grafana

AI: LangGraph, Google ADK, RAG, Agentic Workflows

EDUCATION

University of Illinois at Chicago

Bachelor of Science in Management Information Systems

Dec 2022